

A High-Dynamic-Range Mesh Screen VR Display by Combining Frontal Projection and Retinal Projection

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ABSTRACT

We proposed a high-dynamic-range virtual reality (VR) display that can represent a glossy material by combining a conventional frontal projection mapping and a retinal projection light passing through a screen of a microporous plate. In the prototype system, the retinal projection could be superimposed on the frontal projection and increase the luminance of only the specular highlight in the image. This paper also reports approximately six hundred times brighter presentation of the retinal projection than that of the frontal projection.

Index Terms: Human-centered computing—Visualization—Visualization systems and tools

1 INTRODUCTION

Projection mapping is a technology that changes the appearance of a real object by superimposing an image of a virtual object. There are many studies that use projection mapping to display desired material textures. For example, Takezawa et al. [1] developed an automobile interior design system using 3D projection mapping. This system can display surface textures, such as the glossiness of automobile dashboards. Texture representation by projection mapping has some challenges, one of which is to display the specular high-lights of a glossy material like polished metal.

The glossy material appearance can be divided into two parts: a low-luminance part (diffuse reflection part) and a locally high-luminance part (specular highlight part). We previously measured the luminance ratio of specular highlights to diffuse reflections and found that, generally, it was several dozen times higher than that of diffuse reflections. Hence, when displaying specular highlight in normal projection mapping, the maximum luminance must be significantly increased.

Although the peak luminance can be increased by using a high-power projector, the high-luminance areas are small and expanding the maximum luminance is meaningless for the majority of the area. Therefore, it is desirable to present only the high-luminance area with another device without using a high-power projector.

2 PROPOSED METHOD

2.1 Optical configuration

In this study, we propose a projection mapping system that combines frontal projection and retinal projection by using a mesh screen in order to express localized high-luminance areas, such as specular highlights. The optical design of the proposed method is shown in Fig. 1.

From the observer's side, an image is projected onto a mesh screen, and the projected light diffusely reflected on the screen is observed (frontal projection). From the opposite side, a laser scanning

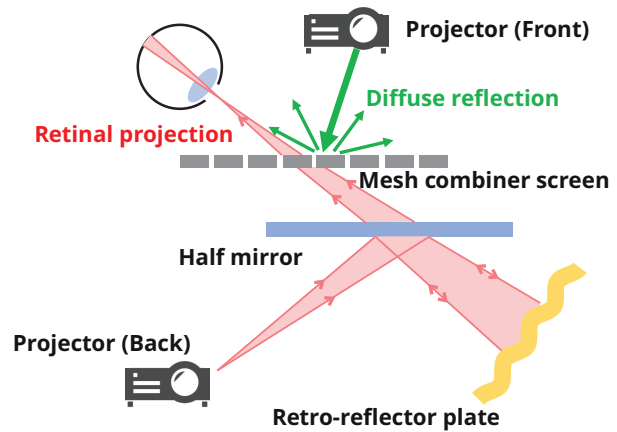


Figure 1: The optical design of the proposed display

projector, a retro-reflector plate, and a half mirror are used to project the image directly onto the retina, using laser light passing through the mesh screen (retinal projection). The light emitted from the laser scanning projector travels along the path shown in red in Fig. 1, is focused on a single point at a position that is plane-symmetrical to the projector with respect to the laser half mirror. The laser light passes through the center of the pupil, and the image is projected directly onto the retina. Since retinal projection is focus-free and much the projected light can enter the pupil, retinal projection is efficient to represent high luminance than diffuse reflection.

When viewing light emitted directly from the display pixels of a liquid crystal display and projected light on a screen, the light entering the pupil is only a small portion of the total diffused light flux. However, when light emitted from a projector is projected onto the retina, all the light from the projector can enter the pupil. Therefore, retinal projection is very efficient in transmitting the light emitted from the source until it reaches the retina, and can express high luminance more easily than diffuse reflection.

2.2 Implementation

Figure 2 shows the prototype system that consists of a frontal projector (focus-free laser scanning type, SK Telecom, LB-UH6CB, 100 lm) and a retinal projector (focus-free laser scanning type, SONY, MP-CL1, 37 lm).

The screen is a stainless steel mesh plate. The diameter of the stainless wire of the mesh plate is 0.06 mm, and it has 150 mesh per inch. The percentage of light passing through the screen is 42%. The mesh plate is made of stainless steel wires woven in a grid pattern, as shown in Fig. 3(a). Fig. 3(b) is 3D model representing the mesh plate's structure. Light incident from the frontal side is diffusely reflected by the mesh stainless steel wire.

A mesh screen has many gaps, and is made of stainless steel wires that are less diffuse-reflective than a white screen. Therefore, the quality of the projected image will be lower than when an image is projected onto a white screen. In order to verify that mesh plate

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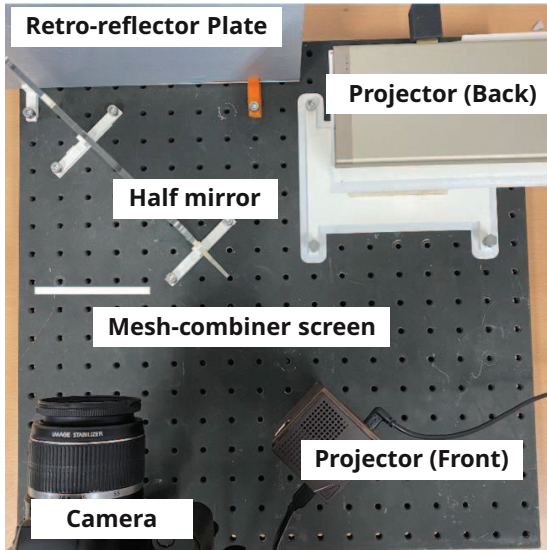


Figure 2: The prototype optics

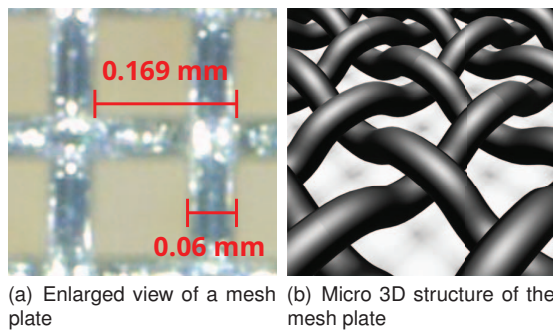


Figure 3: Mesh plate used as a combiner screen

can be used as a combiner screen, we projected natural image onto a white screen (a paper pasted on polystyrene board) and a mesh screen. Fig. 4 shows the results of frontal projection. Although the projected image on the stainless steel mesh screen is uneven and appears to have superimposed granular noise, the projected image is still visible.

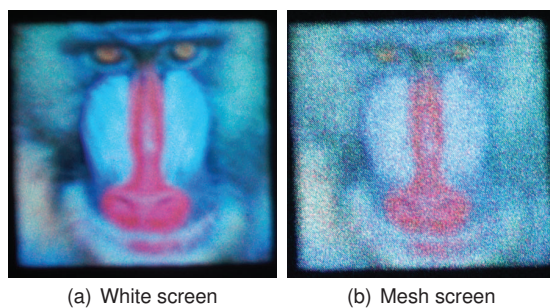


Figure 4: Appearance of frontal projection

3 SUPERIMPOSING RETINAL PROJECTION ON FRONTAL PROJECTION

We experimented using the prototype system to display an image with specular highlights. In frontal projection, the image shown in Fig. 5(a) was projected. In retinal projection, only the highlights were extracted and the image shown in Fig. 5(b) was projected.

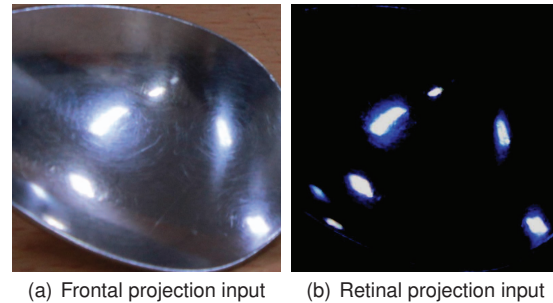


Figure 5: Input images for two types of projectors

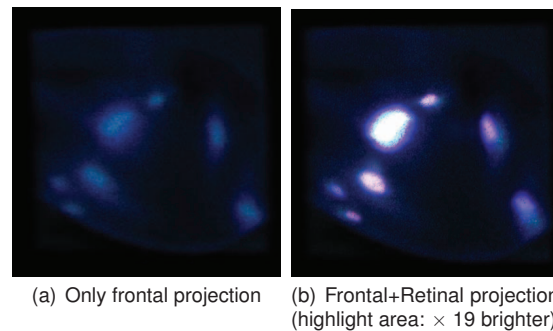


Figure 6: Appearance of combined projection

The result is shown in Fig. 6. The retinal projection could be superimposed on the frontal projection, and retinal projection increased the luminance of only the specular highlight in the image.

In order to measure the luminance ratio between frontal projection and retinal projection, an experiment was conducted using the prototype system and it was found that the retinal projection was about six hundred times brighter than the frontal projection.

4 CONCLUSION

In this study, we proposed a projection mapping system in which a retinal projection is superimposed on a frontal projection by using a stainless steel mesh plate as a combiner screen in order to increase the feasibility of practical implementation. The prototype system can increase the luminance of only the specular highlight and has a retinal projection luminance approximately 19 times brighter than that of the frontal projection, which is sufficient to represent specular highlights such as on metallic surfaces.

REFERENCES

- [1] T. Takezawa, D. Iwai, K. Sato, T. Hara, Y. Takeda, and K. Murase. Material surface reproduction and perceptual deformation with projection mapping for car interior design. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 251–258, 2019. doi: 10.1109/VR.2019.8797923